

Prove each limit using Cauchy's definition of limit.

1. $\lim_{x \rightarrow -3} (-7x - 5) = 16$ (This will give you some practice with some negative values but remember that ε and δ must be positive)

2. $\lim_{x \rightarrow 2} 3x^2 = 12$

3. $\lim_{x \rightarrow 0} x^3 - 4x = 0$ (Be careful here when determining the max value of the other stuff. It does not occur at an endpoint.)